

```

import pandas as pd
from scipy.stats import f_oneway

# =====
# Load Dataset
# =====
file_path = r"C:\Users\Dell\Documents\Modified_Crime_Dataset.csv"
data = pd.read_csv(file_path)

# Recreate 'Total_Crimes' column if it doesn't exist
if 'Total_Crimes' not in data.columns:
    crime_columns = data.columns[3:] # Adjust based on your dataset
    data['Total_Crimes'] = data[crime_columns].sum(axis=1)

# =====
# Group Data by States
# =====
# Get unique states
states = data["STATE/UT"].unique()

# Group data by state and extract 'Total_Crimes'
state_data = [data[data["STATE/UT"] == state]["Total_Crimes"].dropna() for
               state in states]

# =====
# Perform One-Way ANOVA
# =====
f_stat, p_value = f_oneway(*state_data)

# Print ANOVA Results
print("\nANOVA Results:")
print(f"F-Statistic: {f_stat:.3f}")
print(f"P-Value: {p_value:.3f}")

# Decision Making
alpha = 0.05
if p_value < alpha:
    print("Reject the Null Hypothesis: At least one state has a significantly
          different mean total crimes.")
else:
    print("Fail to Reject the Null Hypothesis: No significant difference in
          mean total crimes among the states.")

```

ANOVA Results:

F-Statistic: 7.628

P-Value: 0.000

Reject the Null Hypothesis: At least one state has a significantly different mean total crimes.

